Naive Bayes - Practical Implementation

Virtual Labs - Dayalbagh Educational Institute

Step 1: Importing Libraries

 # Importing necessary libraries

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt
from IPython.display import HTML, display
print("Libraries Imported")
```

Output:

```
Libraries Imported
```

Step 2: Loading Dataset

 # Read the Spam SMS dataset

```
data = pd.read_csv('spam_detection.csv', encoding='latin1', header=0, usecols=[0,1],
    engine="python")
print("Dataset loaded successfully")
```

Output:

```
Dataset loaded successfully
```

Step 3: Data Analysis

 # Display first 5 rows

```
data.head()
```

Output:

```
Category  Message
0  spam      Get your garden re
1  spam      This is your chanc
2  ham       Love is not a deci
3  ham       Good friends care
4  spam      Get your garden re
```

 # Display Last 5 rows

```
data.tail()
```

Output:

```
Category  Message
1737  spam      Do you want 750 an
1738  ham       Thanx...
1739  spam      88800 and 89034 ar
1740  spam      Today's Offer! Cla
1741  ham       am up to my eyes i
```

Displays summary of dataset, including column names, data types, and non-null counts.

```
data.info()
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1742 entries, 0 to 1741
Data columns (total 2 columns):
#   Column    Non-Null Count  Dtype
---  ------    --------------  -----
0   Category  1742 non-null   object
1   Message   1742 non-null   object
dtypes: object(2)
memory usage: 27.3+ KB
```

Checks the dataset for missing(NULL) values in each column

```
data.isnull().sum()
```

Output:

```
Category    0
Message     0
dtype: int64
```

Shows the size of the dataset

```
data.shape
```

Output:

```
(1742, 2)
```

Counts the frequency of each unique category

```
data['Category'].value_counts()
```

Output:

```
Category  Count
Ham       1003
Spam      739
dtype: int64
```

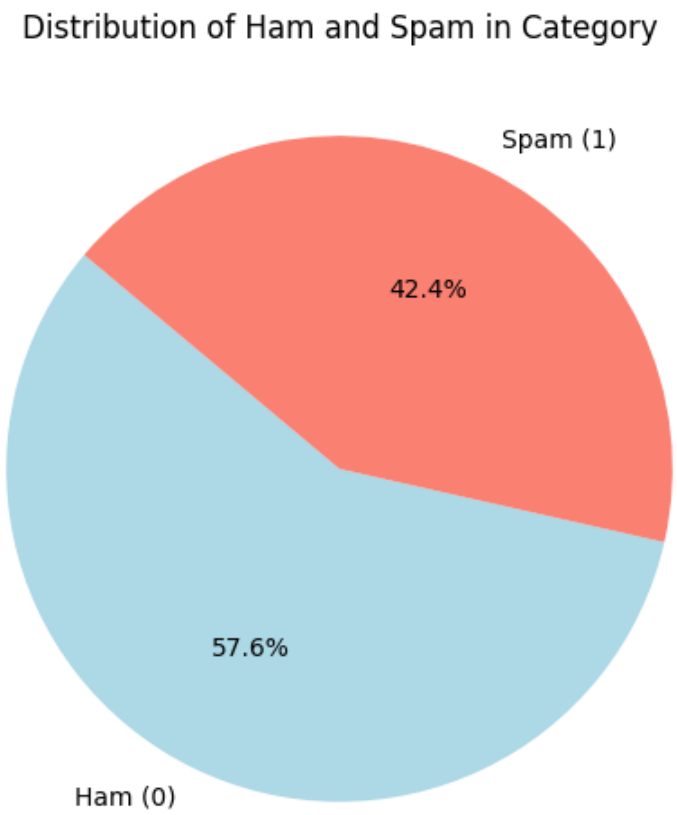
Step 4: Data Preprocessing

 # Visualizes the distribution of spam and ham messages in the dataset

```
category_counts = data['Category'].value_counts()
plt.figure(figsize=(6, 6))
plt.pie(category_counts, labels=['Ham (0)', 'Spam (1)'],
    autopct='%1.1f%%', startangle=140, colors=['lightblue', 'salmon'])
plt.title('Distribution of Ham and Spam in Category');
plt.show()
```

Output:

```
Distribution of Ham and Spam in Category
```



 # Sets up stemming and stop-word removal tools for text preprocessing.

```
stemmer = PorterStemmer()
stop_words = set(ENGLISH_STOP_WORDS)
print("Initialized Porter Stemmer and English stopword list.")
```

Output:

```
Initialized Porter Stemmer and English stopword list.
```

 # Performs basic text cleaning and normalisation

```
def preprocess_text(s):
    if pd.isnull(s): return ""
    s = s.lower(); s = re.sub(r'http\S+|www\.\S+', ' ', s); s = re.sub(r'[^a-z0-9\s]', '
    ', s)
    s = re.sub(r'\s+', ' ', s).strip(); return s
print("Text Cleaned")
```

Output:

```
Text Cleaned
```

Step 5: Model Training

 # Encodes spam labels and splits the data into stratified training and testing sets.

```
data['label'] = (data['Category'].str.lower().str.strip() == 'spam').astype(int)
X_train, X_test, y_train, y_test = train_test_split( data['Message'], data['label'],
    test_size=0.25, random_state=42, stratify=data['label'] )
print("Labels encoded and data split into training and testing sets.")
```

Output:

```
Labels encoded and data split into training and testing sets.
```

 # Converts text messages into TF-IDF feature vectors.

```
vectorizer = TfidfVectorizer(ngram_range=(1,3), min_df=2, max_features=50000)
X_train_vec = vectorizer.fit_transform(X_train)
X_test_vec = vectorizer.transform(X_test)
print("Text data vectorized using TF-IDF.")
```

Output:

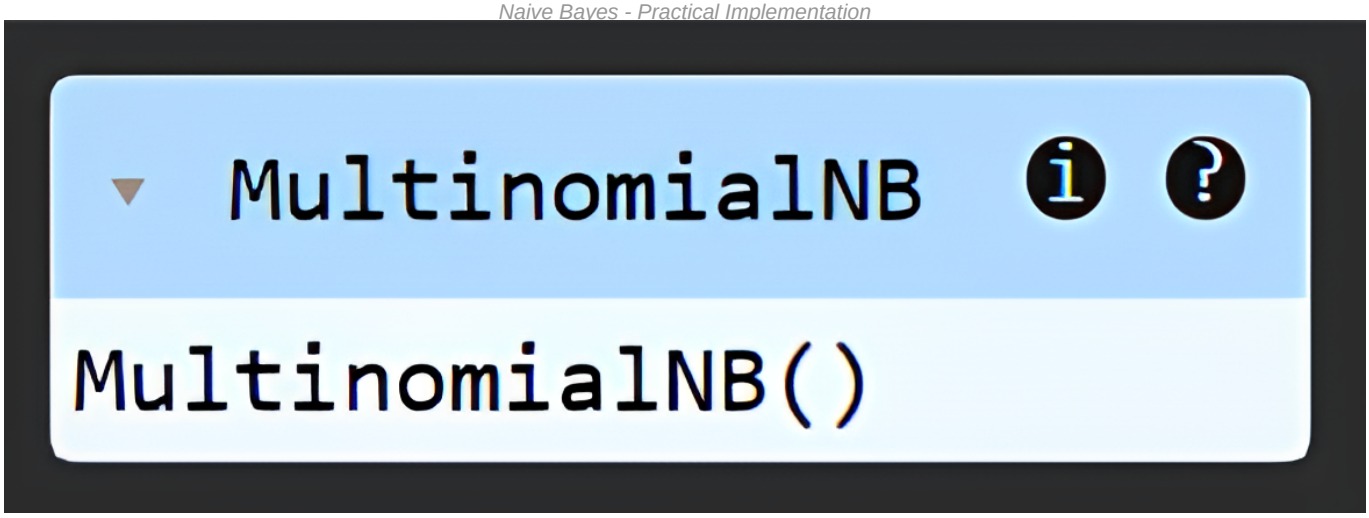
```
Text data vectorized using TF-IDF.
```

 # Trains a Multinomial Naive Bayes classifier on the TF-IDF features.

```
model=MultinomialNB()
model.fit(X_train_vec,y_train)
print("Model training completed.")
```

Output:

```
Model training completed.
```



Step 6: Model Evaluation

 # Predicts on test data and evaluates model

```
y_pred = model.predict(X_test_vec)
y_prob = model.predict_proba(X_test_vec)[:, 1]
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\\nClassification Report:\\n", classification_report(y_test, y_pred))
```

Output:

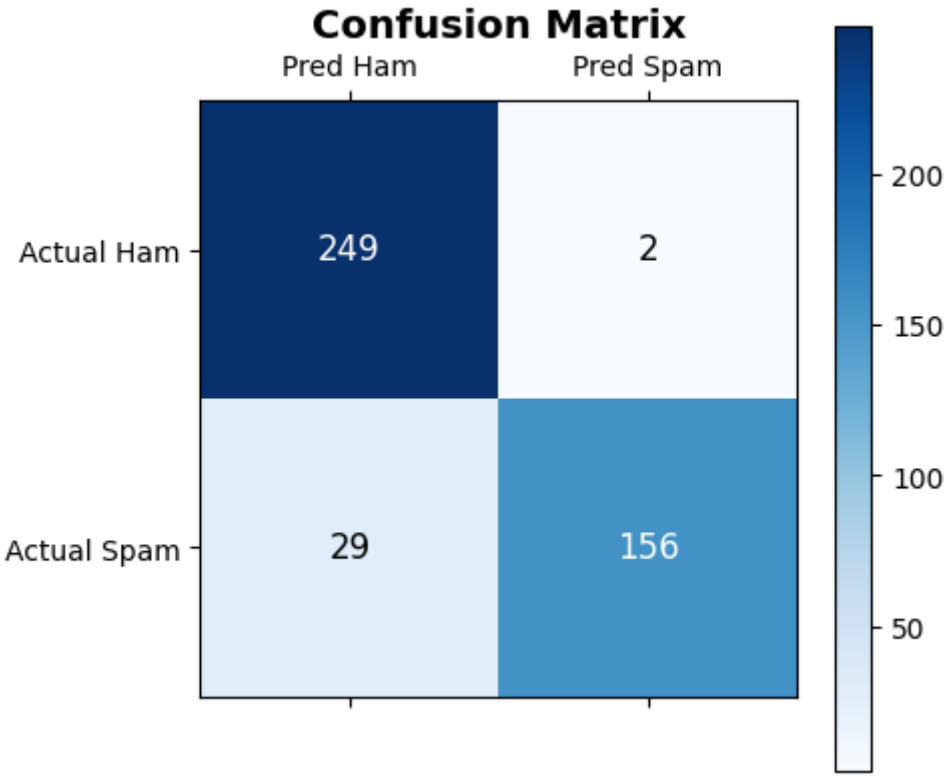
```
Accuracy: 0.9288990825688074
Classification Report:
precision    recall  f1-score   support
0(ham)       0.90      0.99      0.94       251
1(spam)       0.99      0.84      0.91       185
accuracy                           0.93       436
macro avg       0.94      0.92      0.93       436
weighted avg       0.93      0.93      0.93       436
```

 # Plots the confusion matrix with values to visualize correct and incorrect predictions

```
cm = confusion_matrix(y_test, y_pred); plt.matshow(cm, cmap='Blues')
for (i, j), val in np.ndenumerate(cm):
    plt.text(j, i, val, ha='center', va='center', color='white' if val > cm.max()/2 else
    'black', fontsize=12)
plt.xticks([0,1], ['Pred Ham','Pred Spam'])
plt.yticks([0,1], ['Actual Ham','Actual Spam']); plt.colorbar();
plt.show(); print("Confusion matrix displayed.")
```

Output:

```
Confusion matrix displayed.
```

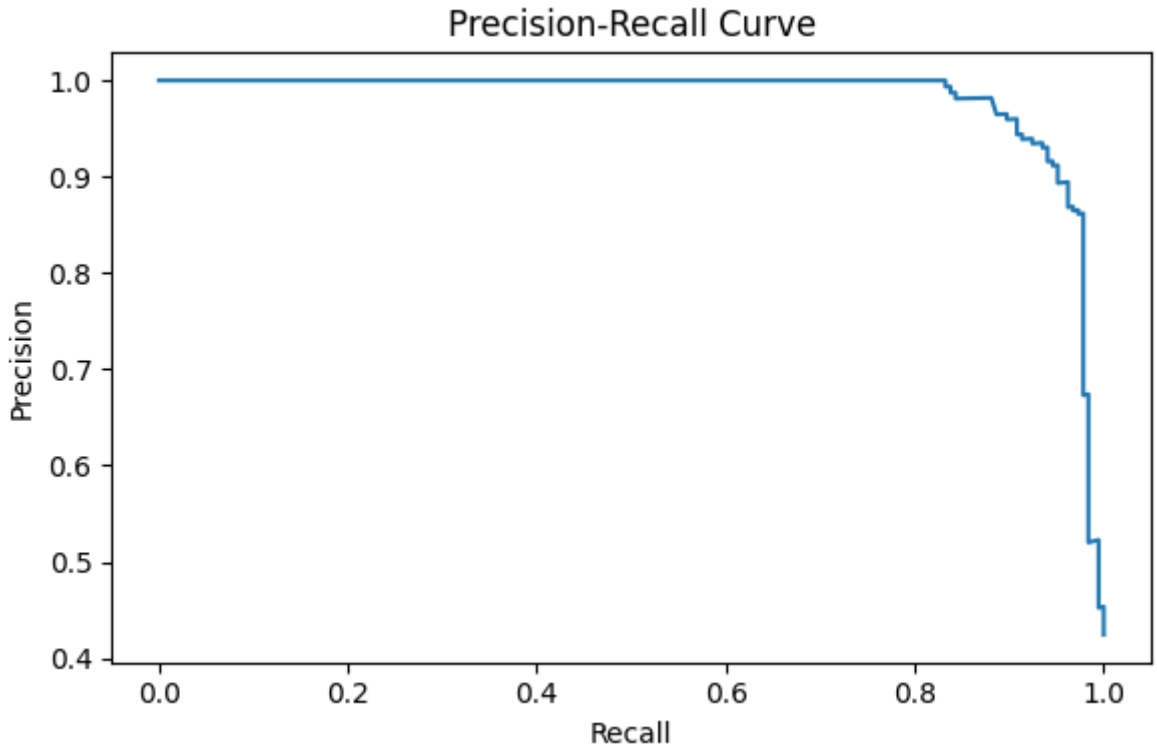


 # Plots the Precision-Recall curve for the model

```
precision, recall, _ = precision_recall_curve(y_test, y_prob)
plt.figure(figsize=(6,4)); plt.plot(recall, precision); plt.title('Precision-Recall
    Curve'); plt.xlabel('Recall');plt.ylabel('Precision'); plt.tight_layout(); plt.show()
print("Precision-Recall curve displayed.")
```

Output:

```
Precision-Recall curve displayed.
```



 # Highlights words based on their contribution to spam/ham prediction using color intensity

```
def color_text(words, contributions):
    max_val = max(abs(c) for c in contributions) or 1; html = ""
    for w, c in zip(words, contributions):
        intensity = int(255 * (abs(c) / max_val))
        if c > 0: color = f"rgb(255,{255-intensity},{255-intensity})"
        else: color = f"rgb({255-intensity},255,{255-intensity})"
        html += f"{w} "
    return html
print("Text contribution visualization function defined.")
```

Output:

```
Text contribution visualization function defined.
```

 # Computes and displays precision and recall of the model on test data

```
def show_precision_recall(model, X_test_vec, y_test):
    preds = model.predict(X_test_vec)
    precision = precision_score(y_test, preds) ;recall = recall_score(y_test, preds)
print("Precision and recall calculation function ready.")
```

Output:

```
Precision and recall calculation function ready.
```

 # Performs precision-recall evaluation and analyzes 'n' random test messages

```
def analyze_multiple(model, vectorizer, X_test, y_test, n=5):
    show_precision_recall(model, X_test_vec, y_test)
    for i in range(n):
        analyze_random_message(model, vectorizer, X_test, y_test)
print("Multiple message analysis function defined.")
```

Output:

```
Multiple message analysis function defined.
```

Runs evaluation and displays analysis of 5 random test messages

```
analyze_multiple(model, vectorizer, X_test, y_test, n=5)
print(" MODEL PERFORMANCE ON TEST DATA")
print("Precision:", round(precision, 4))
print("Recall:", round(recall, 4))
```

Output:

```
MODEL PERFORMANCE ON TEST DATA
Precision: 0.9873
Recall   : 0.8432
Choose one random test sample.
Category  Message
ham       It's getting me do
spam      Text and meet some
ham       I.ll give her once
spam      Congratulations U
ham       thinking about you
Message Analysis
Actual:
Predicted:
Spam Probability:
```